

Information-Theoretic Characterization of Aging Information and Noise in On-Chip Slack Sensors for Silicon Lifecycle Management

Danilo Alencar

Programa de Pós-Graduação em Engenharia de Teleinformática (PGETI)

Universidade Federal do Ceará

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Abstract

On-chip slack sensors used in Silicon Lifecycle Management (SLM) report a quantity that mixes a permanent, slowly-drifting aging component with a reversible component driven by process, voltage and temperature (PVT). Prior work quantifies this contamination with the coefficient of determination (R^2) of a linear regression of slack on temperature and voltage—a measure that captures only linear dependence and yields no transferable unit. This work places an *information-theoretic measurement layer* on top of an existing low-cost accelerated-aging burn-in platform and its synchronized (slack, T , V , time) logs. Treating the sensor as a noisy communication channel, we (i) replace R^2 with the mutual information $I(S; T, V)$ as the correct, non-linear measure of environmental contamination; (ii) quantify, in bits, how much uncertainty PVT logging removes via $H(S) - H(S | T, V)$; (iii) estimate the sensor’s effective resolving power through a channel-capacity heuristic and a Binary Symmetric Channel model of the aging-alarm decision; (iv) derive a noise-limited detectability bound for remaining-useful-life (RUL) estimation, via both a Cramér–Rao/Bayesian point estimate and a sequential Wiener-process/ Kalman-filter treatment that yields a denoised aging trajectory with credible bands and a full, closed-form Inverse-Gaussian RUL distribution. We further apply MDL model selection and ICA-based blind source separation to the same data, and a KL/Jensen–Shannon comparison of two validation benches. On a 214.6 h PID-controlled campaign (768,884 post-soak samples; effective sample size $n_{\text{eff}} = 1,224$) we measure $H(S) = 3.43$ bits and $I(S; T, V) = 1.132$ bits (KSG cross-check 1.221 bits). This measured dependence exceeds the Gaussian-equivalent MI of the prior work’s linear fit ($R^2 = 0.738 \Rightarrow 0.967$ bits) by 26 %, directly evidencing the non-linear environmental coupling that R^2 cannot capture. The contribution is a correct, transferable noise metric for SLM sensors and a principled, optimizable definition of burn-in quality as the minimization of environment–sensor mutual information.

Keywords: information theory; mutual information; entropy; channel capacity; silicon lifecycle management; aging sensor; slack; burn-in; remaining useful life.

1 Introduction

1.1 Motivation and point of departure

Integrated circuits degrade. Bias-temperature instability (BTI/NBTI), hot-carrier injection and related mechanisms slow critical paths over a device’s lifetime, and SLM aims to monitor that degradation in-field to enable adaptive voltage scaling and remaining-useful-life (RUL) prediction. This project builds on three correlated prior efforts: an auto-tuning on-chip slack sensor [1], a thermal-control fidelity study comparing bang-bang and PID burn-in ovens on the same

device and sensor [2], and a low-cost automated burn-in platform [3]. Across all three, a single fact recurs: the slack a sensor reports is a *mixture* of a permanent aging term and a reversible PVT term,

$$S(t) = S_{\text{aged}}(t) - \Delta_{\text{PVT}}(T, V) + \varepsilon_q, \quad (1)$$

where ε_q is quantization noise bounded by the sensor LSB (19.85 ps per count).

1.2 The gap

The prior work measures the reversible contamination with the linear R^2 of slack regressed on T and V (e.g. 0.921 under bang-bang vs. 0.592 under PID control). This has two limitations central to this physical domain: it captures *only linear* dependence, although BTI is super-linear in V and the Arrhenius acceleration is exponential in inverse temperature; and it yields *no transferable unit*—“variance explained” does not answer how many aging states the sensor resolves, how early a trend is detectable, or how much aging information a given bench preserves. Information theory answers all three in one currency: **bits**.

1.3 Contribution

We add an analytical measurement layer—no new silicon, sensor, or oven—that re-expresses noise, correction and recoverability in the language of entropy, mutual information and channel capacity. Concretely:

1. $I(S; T, V)$ replaces R^2 as a non-linear contamination measure (section 6);
2. an entropy account of PVT correction quantifies, in bits, the uncertainty each auxiliary channel removes;
3. a capacity estimate and a BSC alarm model convert measured noise into resolvable aging states and alarm reliability;
4. a Cramér–Rao/Bayesian detectability bound states the earliest detectable aging trend given the noise floor, complemented by a Wiener-process/Kalman-filter treatment that turns the same problem into a continuously-updated, denoised aging trajectory and a closed-form Inverse-Gaussian RUL *distribution* (section 6.6).

We additionally exercise source coding, Kolmogorov/MDL model selection and ICA-based blind separation on the same logs, so that the project spans the full syllabus of the course (section 4 and the topic mapping).

1.4 Scope and honesty

The primary dataset is a single 214.6 h PID campaign on one device; it is long enough to expose a measurable aging drift but cannot, alone, cross-validate a permanent-aging trajectory across devices. RUL results are thus framed as a *methodology and a detectability bound*. Mutual-information estimation on strongly autocorrelated series is bias-prone; we report everything on an effective sample size n_{eff} with block-bootstrap intervals.

2 Background and Related Work

2.1 The on-chip slack sensor

The auto-tuning aging sensor [1] measures critical-path timing slack on a Xilinx Artix-7 FPGA using a dynamically phase-shifted clock, reporting one slack count per 19.85 ps phase increment, and tracks delay degradation under burn-in, temperature and voltage stress.

2.2 Thermal-control fidelity

The thermal-fidelity study [2] compares a thermostatic bang-bang oven against a closed-loop PID oven on the *same* device and sensor, showing that the burn-in thermal architecture changes both the delivered acceleration factor and the reversible environmental noise on the sensor output. It reports a steady-state thermal standard deviation of $\sigma = 0.40$ °C (PID) vs. 3.29 °C (bang-bang), residual slack noise of 0.73 vs. 0.90 counts, and a linear variance-explained of 0.592 vs. 0.921. These are the numbers this work re-expresses in information-theoretic terms.

2.3 The automated burn-in platform

The platform [3] delivers PID-controlled, multi-threaded acquisition logging slack with synchronized temperature and voltage at 1 Hz. The dataset analyzed here (section 5) is one such log.

2.4 Why information theory

Equation (1) is, in information terms, a noisy channel: the transmitter is the latent degradation state, the channel adds reversible PVT fluctuation and quantization noise, and the receiver estimates the slow aging trajectory. This framing makes entropy, mutual information and capacity the natural figures of merit and motivates the methodology of section 4.

2.5 Why R^2 is structurally blind to the physics

The deeper reason R^2 understates environmental coupling is not incidental to this dataset—it is structural. R^2 is a monotone function of the Pearson correlation/covariance between S and its linear fit on (T, V) , and $\text{Cov}(X, Y) = 0$ characterizes only the *absence of linear* association: independence implies zero covariance, but the converse is false in general (a textbook example is $Y = X^2$ for symmetric X , where $\text{Cov} = 0$ yet Y is a deterministic, fully dependent function of X). Consequently any regression- R^2 metric can read as “mostly independent” a variable that is, in fact, entirely determined by the inputs through a non-linear map—precisely the situation here, since BTI aging acceleration is super-linear in V and the Arrhenius rate is exponential in $1/T$. Mutual information (section 3.2) is built from the full joint distribution rather than a second-moment summary, so $I(S; T, V) = 0$ if and only if S and (T, V) are independent, with no linearity assumption—closing exactly the gap this section identifies.

3 Information-Theoretic Framework

3.1 Self-information and entropy

For a discrete source with alphabet $\{s_k\}$ and probabilities $p_k = P(S = s_k)$, the *self-information* of an outcome is $I(s_k) = -\log_2 p_k$: rare outcomes are more surprising and carry more information, and independence makes self-information additive, $I(s_k s_j) = I(s_k) + I(s_j)$ for independent s_k, s_j —the property that forces the logarithm in the definition, since only log turns a product of probabilities into a sum. **Entropy** is the expected self-information,

$$H(S) = -\sum_k p_k \log_2 p_k = E\{I(S)\}, \quad (2)$$

the total uncertainty of the sensor reading, bounded by $0 \leq H(S) \leq \log_2 K$ (K = number of slack levels): zero when one level is certain, maximal when all K levels are equiprobable. The conditional entropy $H(S | T, V)$ is the residual uncertainty once the environment is known, and the chain rule $H(S, T, V) = H(T, V) + H(S | T, V)$ decomposes total uncertainty into

an environmental part and a residual part—the gap between $H(S)$ and $H(S | T, V)$ is the environmental information, formalized as mutual information next.

3.2 Kullback–Leibler divergence and mutual information

For two distributions p, g over the same alphabet,

$$D_{\text{KL}}(p \| g) = \sum_x p(x) \log_2 \frac{p(x)}{g(x)} = E_p \left\{ \log_2 \frac{p(x)}{g(x)} \right\} \quad (3)$$

measures the cost, in bits, of assuming g when the true law is p . $D_{\text{KL}}(p \| g) \geq 0$ with equality iff $p = g$: writing $-\log_2(\cdot)$, a convex function, Jensen’s inequality gives $D_{\text{KL}}(p \| g) = E_p \{ -\log_2 \frac{g(x)}{p(x)} \} \geq -\log_2 E_p \{ \frac{g(x)}{p(x)} \} = -\log_2 \sum_x g(x) = 0$. D_{KL} is *not symmetric* and does not satisfy the triangle inequality, so it is not a metric—exactly the asymmetry exploited in section 6.3 to read $D_{\text{KL}}(p_{\text{SBCCI}} \| p_{\text{PID}})$ as “the cost of describing the noisy bench with the tight reference” and not the reverse. **Mutual information** is the KL divergence between the joint law and the product of marginals,

$$I(S; T, V) = D_{\text{KL}}(p(S, T, V) \| p(S)p(T, V)) = H(S) - H(S | T, V) \geq 0, \quad (4)$$

zero iff $S \perp (T, V)$, and, via the entropy Venn diagram (the union of the $H(S)$ and $H(T, V)$ circles is the joint entropy, their intersection is $I(S; T, V)$), symmetric: $I(S; T, V) = H(T, V) - H(T, V | S)$. Unlike R^2 , eq. (4) captures *arbitrary* dependence, linear or not, because it is built from the full joint law rather than from a second-moment (linear) summary; this is the formal reason a vanishing correlation does not imply independence (section 2) while $I(\cdot) = 0$ does. The normalized *information fraction* $I(S; T, V) / H(S)$ is the information-theoretic analogue of R^2 .

3.3 Differential entropy: why the sensor is treated as discrete

For a continuous variable X with density f , differential entropy $H(X) = -\int f(x) \log_2 f(x) dx$ generalizes eq. (2), but loses two properties that hold in the discrete case: it is not scale-invariant, $H(cX) = H(X) + \log_2 |c|$, and it can be *negative*. Among all densities of fixed variance σ^2 , the Gaussian uniquely maximizes differential entropy, $H(X) = \frac{1}{2} \log_2(2\pi e \sigma^2)$ —a fact used twice in this paper: it is why the Gaussian-equivalent MI conversion of R^2 in section 6.1 is a meaningful comparison, and it is the crux step of the Gaussian channel-capacity derivation below. Because the sensor’s slack output is in reality a 19.85-ps phase count—an integer, not a real number—all the entropy and mutual-information quantities reported for S in this paper (section 6) use the *discrete* definition eq. (2) on the native integer levels, sidestepping the scale-dependence of the differential form and any arbitrary choice of binning.

3.4 Source coding: the entropy as a compression limit

For a prefix code assigning codeword length l_k to symbol k , the Kraft–McMillan inequality $\sum_k 2^{-l_k} \leq 1$ is necessary and sufficient for a decodable prefix code with those lengths to exist, and it implies Shannon’s *source coding theorem*:

$$H(A) \leq L < H(A) + 1, \quad L = \sum_k p_k l_k, \quad (5)$$

so no lossless code can average fewer than $H(A)$ bits per symbol, and a code within one bit of that limit always exists (Huffman achieves it up to rounding). Extending to length- n blocks of an i.i.d. source, $H(A^n) = nH(A)$, so $L_n/n \rightarrow H(A)$: coding longer blocks squeezes the one-bit slack away at the cost of decoding complexity. This paper uses eq. (5) only as the *limit*—“no code can beat $H(S)$ bits/sample”—never instantiating a concrete codec (Huffman/LZW), a deliberate scope decision (section 7) because a codec would be redundant with the entropy and autocorrelation results already computed.

3.5 Channel capacity

For a channel with conditional law $p(y | x)$, capacity is the input distribution that maximizes the mutual information it induces,

$$C = \max_{p_X(x)} I(X; Y), \quad (6)$$

a property of the channel alone (concave in p_X , so the maximum is unique and well defined). Shannon’s *channel coding theorem* states that reliable communication (error probability $\rightarrow 0$) is possible at any rate $r < C$ and *impossible* at any rate $r > C$ —an existence result proved by random coding, not a construction; finding codes that approach C in practice (Turbo, LDPC) took decades. For the binary aging-alarm decision, $X, Y \in \{0, 1\}$ with crossover (error) probability p gives the **Binary Symmetric Channel**

$$C_{\text{BSC}} = 1 - H_2(p), \quad H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p), \quad (7)$$

maximal (1 bit) at $p = 0$ and zero at $p = 0.5$ (the alarm is indistinguishable from a coin flip). For a channel with additive noise, $Y = X + N$ with $X \perp N$, $I(X; Y) = H(Y) - H(N)$; since $H(N)$ does not depend on how X is coded, maximizing $I(X; Y)$ reduces to maximizing $H(Y)$, and by section 3.3 that maximum (for fixed output power) is attained when Y is Gaussian, forcing the capacity-achieving X to be Gaussian as well. With signal power σ_{sig}^2 and noise power σ_{noise}^2 this yields the familiar

$$C \approx \frac{1}{2} \log_2 \left(1 + \frac{\sigma_{\text{sig}}^2}{\sigma_{\text{noise}}^2} \right) \text{ bits/sample}, \quad (8)$$

the discrete-time form of the Shannon–Hartley law. Section 6.4 uses eq. (8) as an *effective-resolution heuristic*—the number of distinguishable aging states the post-correction noise floor permits—not a literal coded-transmission claim: there is no encoder inserting redundancy into the aging process, only a single noisy read-out, so eq. (8) is applied as an upper bound on distinguishability rather than an achievable rate.

3.6 Description-length model selection and Kolmogorov complexity

Shannon entropy quantifies information via the *probability* of an event; Kolmogorov’s independent, probability-free definition [4] quantifies the information in a specific object x as the length of the shortest program that outputs it,

$$K(x) = \min_{p: U(p)=x} l(p), \quad (9)$$

for a fixed universal interpreter U (the invariance theorem guarantees this is well defined up to an additive, x -independent constant across choices of U). This gives Occam’s razor a precise form: among descriptions that fit the data, prefer the one of least description length. The *Minimum Description Length* (MDL) principle [5] operationalizes eq. (9) for statistical model selection, replacing the uncomputable $K(x)$ by an explicit two-part code length $L(\text{model}) + L(\text{data} | \text{model})$: pay for the model’s parameters, then pay to describe the residual under that model. Concretely, for the PVT correction (section 6.2) we use the BIC-style penalty

$$\text{DL} = n_{\text{eff}} \ln \sigma_{\text{resid}}^2 + k \ln n_{\text{eff}}, \quad (10)$$

the (Gaussian, large- n_{eff}) asymptotic form of that two-part code: the first term is the description length of the residual under a Gaussian model of variance σ_{resid}^2 , the second is the cost of the k extra parameters the non-linear correction adds. The non-linear model is adopted only if it compresses the residual enough to pay for its own parameters—exactly the Kolmogorov-complexity intuition that the “best” model is the shortest one that still explains the data, applied here to choose between a linear and a physics-motivated non-linear $\Delta_{\text{PVT}}^{\hat{}}(T, V)$.

3.7 Estimation theory: least squares, MLE, Bayes, Cramér–Rao

PVT corrections are fit by three estimator families, in increasing order of assumption: method of moments / ordinary least squares (linear, distribution-free); maximum likelihood (MLE, non-linear Arrhenius/super-linear form, efficient and consistent under correct specification); and a Bayesian variant carrying a full posterior rather than a point estimate. Posing RUL as estimating a latent aging slope β from noisy observations, the **Cramér–Rao bound** lower-bounds any unbiased estimator’s variance by the inverse Fisher information, $\text{Var}(\hat{\beta}) \geq \mathcal{I}_F(\beta)^{-1}$, which for i.i.d. Gaussian residuals of variance σ^2 observed at n_{eff} effectively independent points reduces to a closed form in σ^2 and n_{eff} (section 6.5); it is the frequentist floor beneath which no estimator—however clever—can pin down the aging rate, given the measured noise and effective sample size. The Bayesian variant complements it with a Student- t posterior (Jeffreys prior on the thinned, i.i.d. series) and a credible interval that is compared against the CRLB point estimate as a cross-check (section 6.5).

3.8 ICA, negentropy and blind source separation

Independent Component Analysis seeks a linear unmixing $\vec{z} = W\vec{x}$ of observed channels $\vec{x} = (S, T, V)$ into components that are maximally statistically independent and non-Gaussian, using **negentropy**

$$N_G(X) = H(X_{\text{Gauss}}) - H(X), \quad (11)$$

as the non-Gaussianity contrast: since the Gaussian maximizes differential entropy at fixed variance (section 3.3), $N_G(X) \geq 0$ with equality iff X is Gaussian, so maximizing an estimate of $N_G(X)$ (via higher-order statistics such as excess kurtosis, as used here, or a Gram–Charlier expansion around the Gaussian) drives the unmixing toward independent, physically distinct sources without needing a parametric model for $\Delta_{\text{PVT}}(T, V)$. Applied to (S, T, V) , ICA tests whether the aging source can be recovered by non-Gaussianity/independence alone, as a model-free counterpart to the regression-based correction of section 3.6; separation quality is scored by the residual inter-component mutual information (ideal value 0), using eq. (4) as the independence measure.

3.9 Sequential estimation: Wiener-process degradation and Kalman filtering

Section 3.7 treats aging-slope estimation as a single, campaign-wide point-estimation problem: one β , fit once, with a Cramér–Rao or Bayesian error bar attached after the fact. The course’s stochastic-process material — Markov chains as the discrete-state, discrete-time model of a system whose future depends on the present alone, not the past — has a direct continuous-state, continuous-time analogue: the **Wiener process**, whose independent, stationary Gaussian increments make it the simplest Markov process with continuous sample paths. Modelling the latent, denoised aging level $X(t)$ (what eq. (1) calls $S_{\text{aged}}(t)$, net of the removable Δ_{PVT} term) as a Wiener process with drift,

$$dX(t) = \mu dt + \sigma_B dW(t), \quad S_{\text{corr}}(t_k) = X(t_k) + v_k, \quad v_k \sim \mathcal{N}(0, R), \quad (12)$$

turns RUL from a single-slope bound into two richer questions: (i) what is the *best current estimate* of $X(t)$, continuously updated and with honestly propagated uncertainty, rather than a single after-the-fact regression line; and (ii) what is the *full probability distribution* of the time to a future degradation milestone, not just a point rate and its standard error. Equation (12) is a linear-Gaussian state-space model, so both are answered by the **Kalman filter** [6]: given a Gaussian prior $X_{k-1} \sim \mathcal{N}(\hat{x}_{k-1}, P_{k-1})$, the recursive predict/update pair

$$\hat{x}_k^- = \hat{x}_{k-1} + \mu \Delta t_k, \quad P_k^- = P_{k-1} + \sigma_B^2 \Delta t_k; \quad \hat{x}_k = \hat{x}_k^- + K_k(y_k - \hat{x}_k^-), \quad K_k = \frac{P_k^-}{P_k^- + R} \quad (13)$$

is the exact Bayesian posterior update at each step (Gaussian priors and Gaussian likelihoods stay Gaussian), and a backward Rauch–Tung–Striebel pass turns the causal filtered estimate into a smoothed, whole-campaign trajectory $\hat{X}(t)$ with credible bands — the direct denoising counterpart to the per-sample S_{corr} residual of section 3.6, now propagated as a posterior over the entire latent path rather than a single detrended series. Letting the drift itself follow a slow random walk, $d\mu(t) = \sigma_{\text{rate}} dW'(t)$ (a *local linear trend* model [7]), makes the constant-rate assumption of section 3.7 a special case ($\sigma_{\text{rate}} \rightarrow 0$) rather than an assumption baked into the model, and its process-noise variance σ_{rate}^2 is fit, like any other model parameter in this paper, by maximizing the Gaussian likelihood of the filter’s own one-step-ahead prediction errors (the innovations form of eq. (13)) — the same maximum-likelihood principle of section 3.7, applied here to a dynamic rather than a static model.

For a Wiener process with drift $\mu < 0$ and diffusion σ_B^2 , the time to first travel a distance D (e.g. one further noise-floor-sized or signal-sized excursion) is classical in accelerated-degradation modelling [8, 9]: it follows an **Inverse Gaussian** distribution,

$$T_D \sim \text{IG}(D/|\mu|, D^2/\sigma_B^2), \quad f(t) = \sqrt{\frac{D^2}{2\pi\sigma_B^2 t^3}} \exp\left(-\frac{(D + \mu t)^2}{2\sigma_B^2 t}\right), \quad (14)$$

with mean $D/|\mu|$ and, notably, variance $D^3/(|\mu|^3\sigma_B^2)$: unlike the Cramér–Rao bound’s single standard error, eq. (14) gives the *entire* RUL distribution in closed form, including its skew — and, because no calibrated failure threshold D is established for this sensor (section 3.7’s honesty caveat), section 6.6 reports it parametrically at thresholds that are already-computed quantities from section 6.4 rather than a new, uncalibrated assumption.

4 Methodology

4.1 Stage 1 — Acquire

Reuse the synchronized (S, T, V, t) logs (section 5); exclude the first 1.0 h thermal-soak transient; retain raw resolution (discretization is deferred to the estimation step).

4.2 Stage 2 — Transform

Detrend the slow aging drift with a robust estimator (long-window median / Theil–Sen) so that $I(S; T, V)$ measures reversible coupling rather than a shared time trend; standardize T, V . Handle serial correlation via (i) the integrated autocorrelation time τ_{int} and an effective sample size $n_{\text{eff}} = n/(2\tau_{\text{int}})$; (ii) moving-block bootstrap confidence intervals; (iii) thinning as a cross-check. Two estimator families are compared: binned plug-in with Miller–Madow bias correction, and k -nearest-neighbour (Kozachenko–Leonenko / KSG); their agreement is a robustness check.

4.3 Stage 3 — Quantify

Estimate $H(S)$, $H(S | T, V)$, the per-channel $I(S; T)$, $I(S; V)$, the joint $I(S; T, V)$ and the information fraction; compare regimes; compute $D_{\text{KL}}(\cdot \| \cdot)$ between benches (eq. (4)).

4.4 Stage 4 — Preserve (PVT correction as information recovery)

Fit $\hat{\Delta}_{\text{PVT}}(T, V)$ (linear OLS; non-linear GLS/MLE; Bayesian) and form $S_{\text{corr}} = S - \hat{\Delta}_{\text{PVT}}$. Judge each correction by the *reversible* (detrended) residual MI $I(S_{\text{corr}}; T, V) \rightarrow 0$ and the residual standard deviation relative to the quantization floor; select the model by MDL. Rank T vs. V by information contribution (feature selection). As an exploratory dual, run ICA/negentropy blind separation and score it by residual inter-component MI.

4.5 Stage 5 — Infer (capacity and RUL)

Convert post-correction residual noise into resolvable aging states via eq. (8); model the threshold alarm as a BSC and report $C_{\text{BSC}} = 1 - H_2(p)$ (eq. (7)). Derive the Cramér–Rao detectability bound on the aging slope using n_{eff} , complemented by a Bayesian posterior. As a second, sequential-estimation route to the same question (section 3.9), block-average S_{corr} into $\sim n_{\text{eff}}$ quasi-independent observations (block length $2\tau_{\text{int}}$, i.e. one n_{eff} unit per block by construction), fit the Wiener diffusion σ_B^2 and the block observation noise R by the method of moments (variance of block-mean increments vs. lag), fit the rate-noise σ_{rate}^2 by 1-D MLE on the Kalman filter’s innovation likelihood, and run a Kalman/RTS smoother to obtain a denoised aging trajectory $\hat{X}(t)$ with credible bands and the Inverse-Gaussian RUL distribution of eq. (14) at two already-computed thresholds ($D = \sigma_{\text{noise}}$, $D = \sigma_{\text{signal}}$). Finally, frame “best burn-in condition” as minimizing $I(S; T, V)$ subject to delivering the required Arrhenius acceleration factor.

4.6 Validation

Cross-estimator agreement (binned Miller–Madow vs. KSG); cross-bench ordering consistent with the published σ ranking; phase-randomized surrogate/null tests to calibrate the MI bias floor; dither on/off to control for quantization artifacts.

5 Dataset

The primary substrate is a single PID-controlled accelerated-aging campaign (`Tentative_1_20260524_203756.csv`), started 2026-05-24, sampled at 1 Hz for 214.6 h (772,484 raw rows, of which 5 are link-down dropouts). After excluding the first 1.0 h soak transient, 768,884 samples remain. The DUT is held in a high-stress steady state ($T \approx 125.1^\circ\text{C}$, $V \approx 1.400\text{ V}$) under PID control with an identified FOPDT plant model $G(s) = 1.56 e^{-150.6s} / (1307.2s + 1)$.

Table 1 summarizes the sensor-relevant channels and the measured relationships. Slack occupies 27 integer levels over the range 267–300 counts, giving a discrete entropy of $H(S) = 3.43$ bits. Temperature is the dominant reversible driver ($\text{corr}(S, T) = -0.859$), with a weaker voltage coupling (+0.213) and a genuine slow aging drift ($\text{corr}(S, t) = -0.348$; raw least-squares slope -0.0171 counts/h). A linear regression of slack on (T, V) explains $R^2 = 0.738$ of the variance—a value that, because the underlying physics is non-linear, is a *lower bound* on the true environmental coupling and thus the central motivation for the mutual-information analysis of section 6.

The data hazards carried into the pipeline are: heavy serial correlation ($n_{\text{eff}} \ll n$, with $n_{\text{eff}} = 1,224$ here), integer quantization of S , and a trend/environment confound requiring detrending before MI.

For the bench comparison of section 6.3, two ~ 24 h *stabilized* runs are used: a non-PID SBCCI bench (slack noise 3.23 counts) and the PID-controlled reference (1.46 counts). These were acquired on a **dedicated test device distinct from the 214 h aging device**; the two benches share that test device, so their divergence isolates the bench/environment difference, but their absolute slack levels are not comparable to the aging device’s. Their differing operating points are handled by comparing the centered slack fluctuation, so the divergence reflects noise character rather than setpoint.

6 Results

6.1 Mutual information vs. R^2

Table 2 sets the information measures beside the linear R^2 baseline on the PID campaign. The binned (Miller–Madow) estimate gives $I(S; T, V) = 1.132$ bits, with temperature dominating

Table 1: Sensor-relevant channels and measured relationships on the post-soak window ($n = 768,884$).

Quantity	Value	Reading
Duration	214.6 h	single PID campaign
Post-soak samples	768,884	after 1 h soak
T (mean)	125.1 °C	high-stress steady state
V (mean)	1.400 V	held by PSU
S (mean / std)	278.5 / 3.03	counts
S levels / range	27 / 267–300	occupied integer states
$\text{corr}(S, T)$	-0.859	temperature dominates
$\text{corr}(S, V)$	+0.213	weak, correct sign
$\text{corr}(S, t)$	-0.348	slow aging drift
raw slope dS/dt	-0.0171 cnt/h	confounded w/ PVT
linear $R^2(S T, V)$	0.738	lower bound on coupling
$H(S)$ (discrete)	3.43 bits	total output budget

($I(S; T) = 1.087$ vs. $I(S; V) = 0.113$ bits); the KSG estimator independently yields 1.221 bits, and the moving-block bootstrap 95 % interval on the binned value is [1.091, 1.176] bits. The cross-estimator agreement is the robustness evidence of section 4.

The non-linearity result. The correct, apples-to-apples comparison is *not* the entropy fraction against the variance fraction (different scales), but the measured MI against the *Gaussian-equivalent* MI implied by the linear fit, $-\frac{1}{2} \log_2(1 - R^2) = 0.967$ bits for $R^2 = 0.738$. The measured dependence (1.221 bits) **exceeds** this linear-equivalent value by 26 %: there is genuine environmental information that a linear model cannot see—exactly the gap this project set out to expose, and direct quantitative support for replacing R^2 with mutual information in this domain.

Surrogate null floor. Because MI estimators are biased upward on finite, autocorrelated samples, the measured value must be checked against the floor the estimator produces under genuine independence. Circularly shifting the (T, V) stream against S by large random lags—destroying the cross-dependence while preserving each channel’s marginal *and* autocorrelation—yields a null $I(S; T, V)$ of mean 0.069 bits and 95th percentile 0.097 bits. The measured 1.132 bits is $\approx 12\times$ this floor, so the dependence is overwhelmingly real rather than estimator bias. The same floor frames the post-correction reversible residual (0.046 bits, section 6.2): having driven the leftover coupling down to the same order as the null floor is exactly what “the environment has been removed” means in information terms.

6.2 PVT correction as information recovery

We fit $\hat{\Delta}_{\text{PVT}}(T, V)$ and form $S_{\text{corr}} = S - \hat{\Delta}_{\text{PVT}}$. The ordinary-least-squares (linear) correction drives the linear coupling to zero by construction ($\text{corr}(S_{\text{corr}}, T) = +0.00$, $\text{corr}(S_{\text{corr}}, V) = -0.00$); the methodologically relevant question is how much *reversible* mutual information it removes. Measured on detrended signals—so that the slow aging drift does not masquerade as environmental coupling—the reversible coupling falls from 0.656 bits before correction to 0.046 bits after, i.e. the linear correction removes **93 %** of the reversible environment–sensor information, leaving a residual at the estimator’s noise floor. The reversible residual standard deviation is 0.84 counts, consistent with the prior work’s published PID residual (0.73 counts)—an independent cross-check of the pipeline.

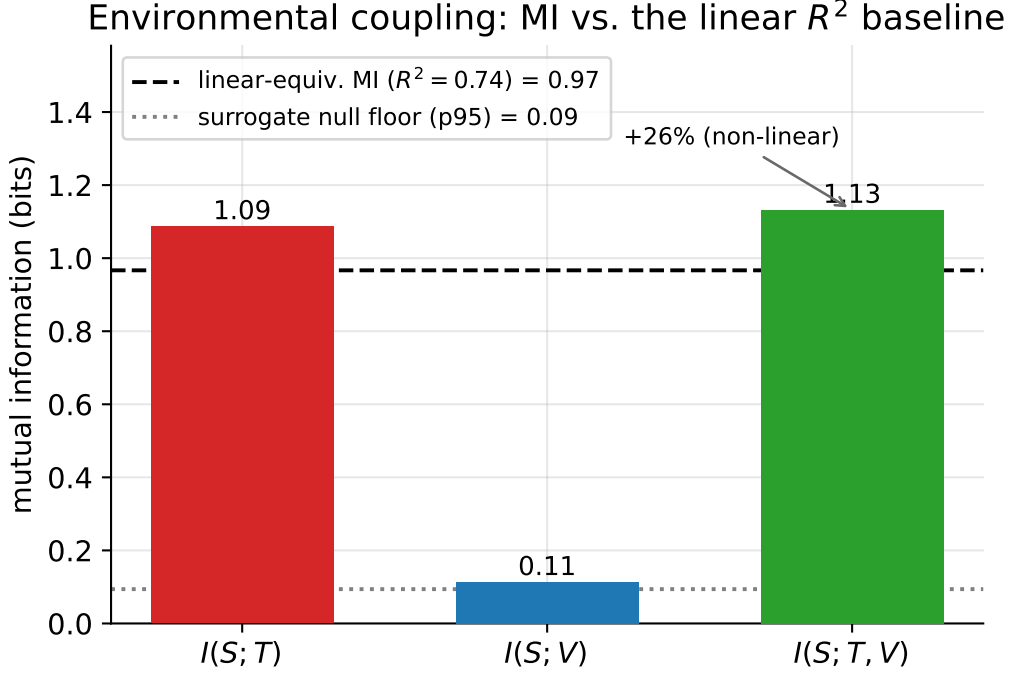


Figure 1: Per-channel and joint mutual information between slack and the environment (bits). The joint $I(S; T, V)$ exceeds the Gaussian-equivalent of the linear R^2 (dashed) by 26%, and sits well above the surrogate null floor (dotted), confirming genuine non-linear coupling.

The physics-motivated non-linear correction (Arrhenius-like in T , super-linear in V) removes marginally more residual MI (0.024 vs. 0.046 bits) but reduces the residual variance negligibly; the MDL/BIC criterion (eq. (10), section 3.6) therefore **selects the linear model** (*linear*), judging the extra parameters unjustified at this stress point—a description-length verdict revisited in section 7.

Aging recovery (key result). Removing the PVT contribution does not merely clean the signal—it *exposes* the latent aging trajectory. The correlation of slack with time strengthens from $\text{corr}(S, t) = -0.35$ on the raw stream to $\text{corr}(S_{\text{corr}}, t) = -0.73$ after correction: stripping the reversible environmental information makes the permanent aging drift the dominant structure in the residual, which is precisely the information-recovery objective of eq. (1) and the empirical anchor for the RUL bound of section 6.5.

6.3 Distributional distance between benches

Two ~ 24 h stabilized runs on different benches—the non-PID SBCCI setup and the PID-controlled reference—let us put a number on “how far the non-compliant environment pushes the slack distribution.” Both runs are on a dedicated test device, distinct from the 214 h aging device of the preceding sections; they share that test device, so the divergence isolates the bench/environment difference rather than device-to-device variation. The benches sit at different operating points, so the comparison is made on the centered slack fluctuation, isolating noise character from setpoint. The slack noise is 3.23 counts (SBCCI) vs. 1.46 counts (PID), tracking the temperature noise (2.64 vs. 1.25 $^{\circ}\text{C}$)—the contamination is thermally driven.

The Kullback–Leibler divergence is strongly *asymmetric*: $D_{\text{KL}}(p_{\text{SBCCI}} \| p_{\text{PID}}) = 1.94$ bits (the cost of describing the noisy bench with the tight PID reference) versus $D_{\text{KL}}(p_{\text{PID}} \| p_{\text{SBCCI}}) = 1.01$ bits—a direct illustration of the course’s point that KL is not a metric. The symmet-

Table 2: Information measures (bits) vs. the linear R^2 baseline on the PID campaign. The measured $I(S; T, V)$ exceeds the Gaussian-equivalent MI of the linear fit ($-\frac{1}{2} \log_2(1 - R^2)$) by 26 %, evidencing non-linear coupling. Prior-work columns are the published thermal-fidelity statistics.

	PID (new run)	PID (prior)	Bang-bang (prior)
linear $R^2(S T, V)$	0.738	0.592	0.921
$\hookrightarrow$ Gaussian-equiv. MI (bits)	0.967	—	—
$H(S)$ (bits)	3.43	—	—
$I(S; T)$ (bits)	1.087	—	—
$I(S; V)$ (bits)	0.113	—	—
$I(S; T, V)$ binned (bits)	1.132	—	—
$\hookrightarrow$ KSG cross-check (bits)	1.221	—	—
$\hookrightarrow$ 95 % CI (block btstrp)	[1.091, 1.176]	—	—
info. fraction $I(;) / H()$	0.330	—	—

ric Jensen–Shannon divergence is 0.279 bits (of a maximum of 1), a single transferable “distance between validation environments.” A zero-mean Gaussian closed form gives 1.67 bits for $D_{\text{KL}}(p_{\text{SBCCI}} \| p_{\text{PID}})$; the empirical value is higher (1.94 bits), and that excess is itself evidence that the non-compliant bench injects *non-Gaussian*, heavy-tailed excursions—which motivates the blind-separation analysis of section 6.7.

6.4 Effective resolution and alarm reliability

Taking the post-correction reversible noise $\sigma_{\text{noise}} = 0.84$ counts and the aging-signal spread $\sigma_{\text{sig}} = 1.29$ counts, the capacity heuristic (eq. (8), derived in section 3.5 from the Gaussian maximum-entropy argument) gives $\text{SNR} = 2.36$, $C = 0.87$ bits and therefore 1.8 distinguishable aging states over this campaign—a deliberately modest figure that reflects an aging excursion only $\sim \sqrt{2.36} \times$ the noise floor, and that the prior residual- σ metric could not express.

The aging-alarm decision is modelled as a BSC whose crossover p follows from the per-window detection SNR (eq. (7)). Reliability is strongly *time-scale dependent*: a per-hour decision is essentially a coin flip ($p \approx 0.49$, $C_{\text{BSC}} \approx 0$) because one hour of drift is buried in the noise, whereas a 24 h operational window gives $p = 0.001$ and $C_{\text{BSC}} = 0.99$ bits/decision (near-perfect), and the full campaign is effectively certain. The crossover between these regimes coincides with the minimum observation time derived next.

6.5 RUL detectability bound

Posing RUL as least-squares estimation of the aging slope, and using the measured noise with the effective sample size $n_{\text{eff}} = 1,224$ ($\tau_{\text{int}} = 314$ samples, i.e. ~ 5 min decorrelation), the Cramér–Rao bound (section 3.7) gives a slope-estimate standard deviation of 0.39 m-cnt/h over the ~ 214 h campaign, hence a minimum detectable aging rate (at 3σ) of **1.2 m-cnt/h**. The measured corrected aging rate is -18.3 m-cnt/h—roughly $15\times$ above this floor, so the trend is firmly detectable. Equivalently, the minimum observation time to resolve the measured trend at 3σ is **~ 34 h**, well within this campaign (and explaining why the prior ~ 23 h runs could not separate aging from the reversible floor). This is the honest detectability statement the project promised: not a lifetime prediction, but a noise-limited bound on *when* aging becomes observable.

Bayesian credible bars. The frequentist bound is complemented by a Bayesian linear regression of S_{corr} on time, fit on the data thinned to the n_{eff} effectively-independent points (so

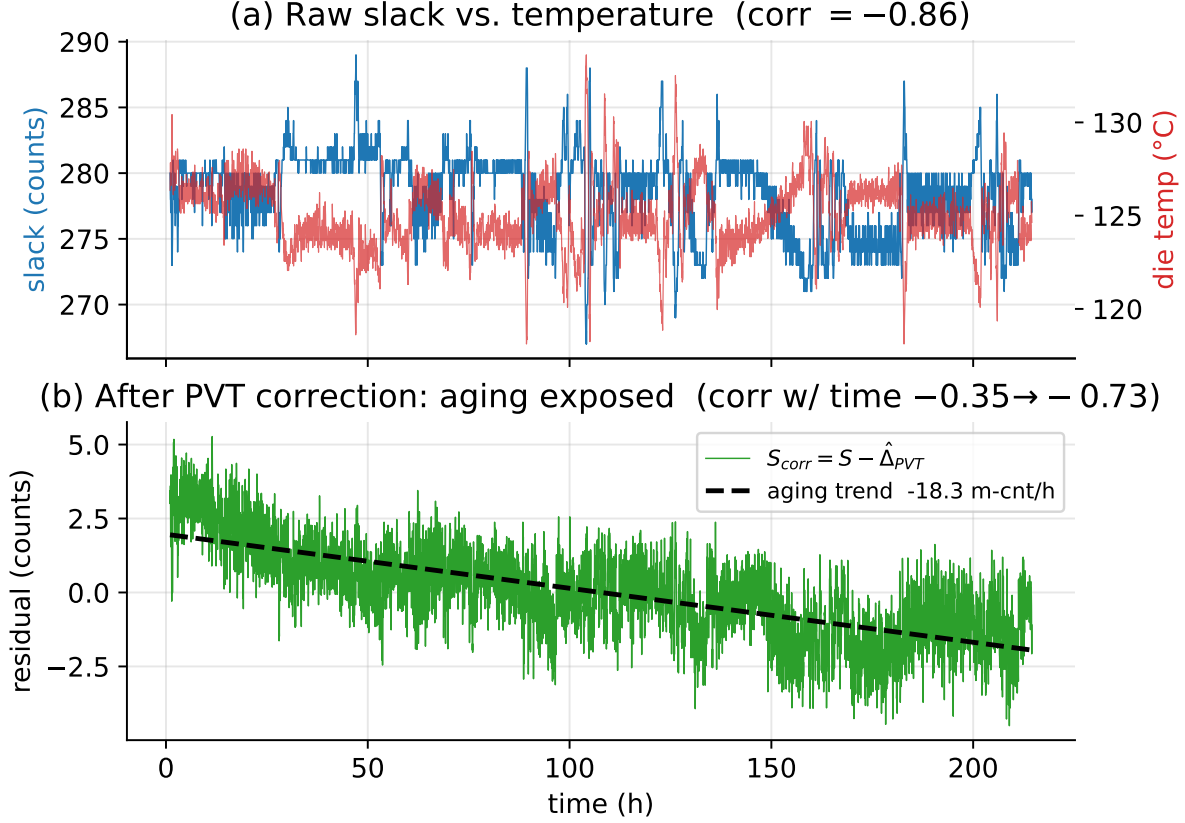


Figure 2: (a) Raw slack tracks die temperature (corr -0.86). (b) After PVT correction the reversible coupling is gone and the permanent aging trend is *exposed* (slack–time correlation strengthens from -0.35 to -0.73); the dashed line is the fitted aging rate.

the i.i.d. likelihood holds) under a Jeffreys prior, giving a Student- t posterior for the slope. The posterior aging rate is -18.2 m-cnt/h with a 95% credible interval of $[-17.2, -19.2]$ m-cnt/h, and the posterior probability that the slope is genuinely negative—that aging is present—is 100.00%, numerically indistinguishable from certainty. Propagating the slope posterior through the detectability relation gives a credible interval of $[33, 36]$ h on the minimum observation time, bracketing the 34 h point estimate. The two paradigms agree, and the Bayesian treatment supplies the calibrated error bars a future RUL study would propagate forward.

6.6 A denoised aging trajectory and RUL as a distribution (Wiener/Kalman)

The Cramér–Rao and Bayesian results above give a single campaign-wide rate and its error bar. Here we ask the sequential-estimation question of section 3.9 instead: block-averaging S_{corr} into 1,224 quasi-independent observations (one per $2\tau_{int}$, i.e. one n_{eff} unit each), the method-of-moments fit of eq. (12) gives a Wiener diffusion $\sigma_B = 0.403$ counts/ $\sqrt{h}$ and a block-mean observation noise R notably *below* the native per-sample σ_{noise}^2 (section 6.4) — direct evidence that averaging over a full decorrelation window still removes sub- τ_{int} structure that the single autocorrelation time alone does not capture. The rate-noise MLE collapses to its numerical floor ($\sigma_{rate}^2 \approx 1.0 \times 10^{-10}$): the data prefer a *constant*-drift Wiener process over a time-varying one, a second, independent confirmation — alongside the MDL verdict of section 6.2 — that this single, narrow-operating-point campaign does not contain enough curvature information to support a more flexible model. The Kalman/RTS-smoothed rate at the end of the campaign is -15.9 m-cnt/h, consistent in sign and order of magnitude with the CRLB/Bayesian estimate of section 6.5 (~ -18 m-cnt/h); the $\sim 15\%$ gap between the two independently-derived rates

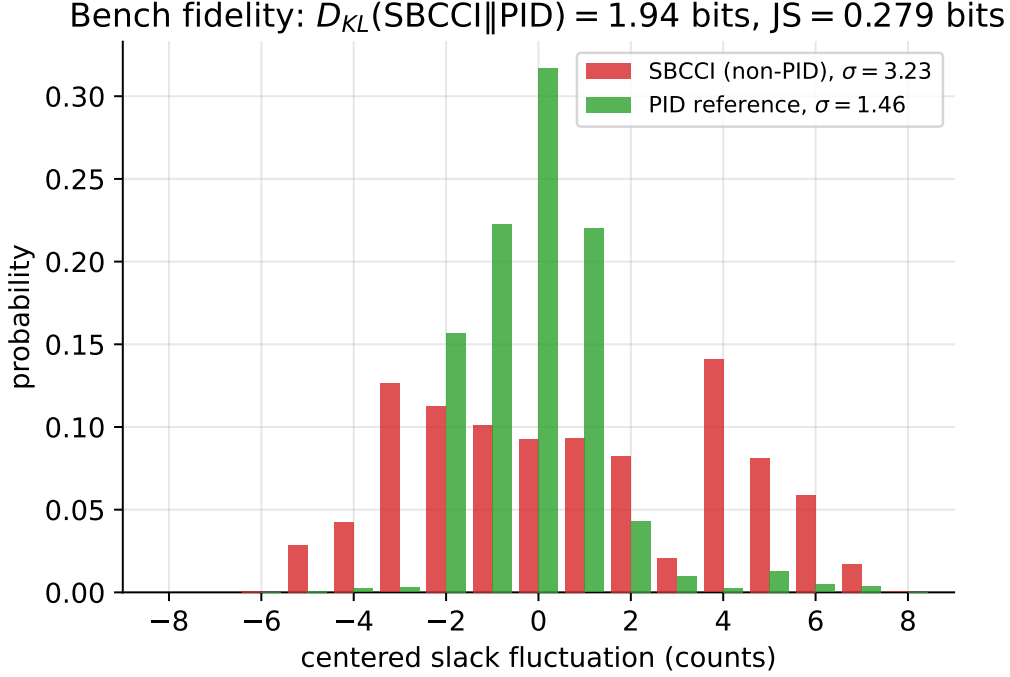


Figure 3: Centered slack-fluctuation distributions of the two benches (same test device). The non-PID SBCCI bench is markedly wider than the PID reference; the divergence $D_{KL}(p_{\text{SBCCI}} \parallel p_{\text{PID}}) = 1.94$ bits ($JS = 0.279$ bits) quantifies the bench-fidelity gap.

reflects the different estimators (whole-series OLS vs. a filtered, block-averaged state), not a contradiction. The smoothed level’s posterior standard deviation is essentially flat across the campaign ($0.214 \rightarrow 0.218$ counts): the filter reaches its steady-state uncertainty within the first few blocks and does not keep shrinking indefinitely, exactly as a linear-Gaussian filter with constant R and σ_B^2 predicts.

RUL as a distribution. At $D = \sigma_{\text{noise}} = 0.84$ counts (the same noise floor that limits the sensor to 1.8 distinguishable states, section 6.4), the Wiener first-passage time has mean 52.9 h but median 8.0 h; at $D = \sigma_{\text{signal}} = 1.29$ counts, mean 81.3 h and median 17.5 h. The large gap between mean and median in both cases is not an artifact: the Inverse Gaussian is strongly right-skewed exactly when the threshold D is small relative to the diffusion scale σ_B , which is precisely the low-SNR ($\text{SNR} = 2.36$, section 6.4) regime measured here — *the same noise limitation that caps the sensor’s resolvable states also makes short-horizon RUL predictions highly uncertain*, a quantitative link between the capacity result and the RUL result that a single point estimate cannot express. This is offered as a *methodology* for a full RUL distribution, exercised at self-referential thresholds because, as throughout this paper, no externally calibrated failure threshold exists for this sensor; supplying one (section 8) turns eq. (14) directly into an operational RUL distribution.

6.7 Blind source separation (ICA)

As a model-free counterpart to the regression correction of section 6.2, FastICA (negentropy contrast, eq. (11) in section 3.8) was applied to the three observed channels (S, T, V) alone, with no functional form assumed for Δ_{PVT} . The recovered component that tracks time matches the regression-based aging residual S_{corr} at $|\text{corr}| = 0.834$ —*model-free ICA independently arrives at essentially the same aging signal the regression produced*, which cross-validates the PVT

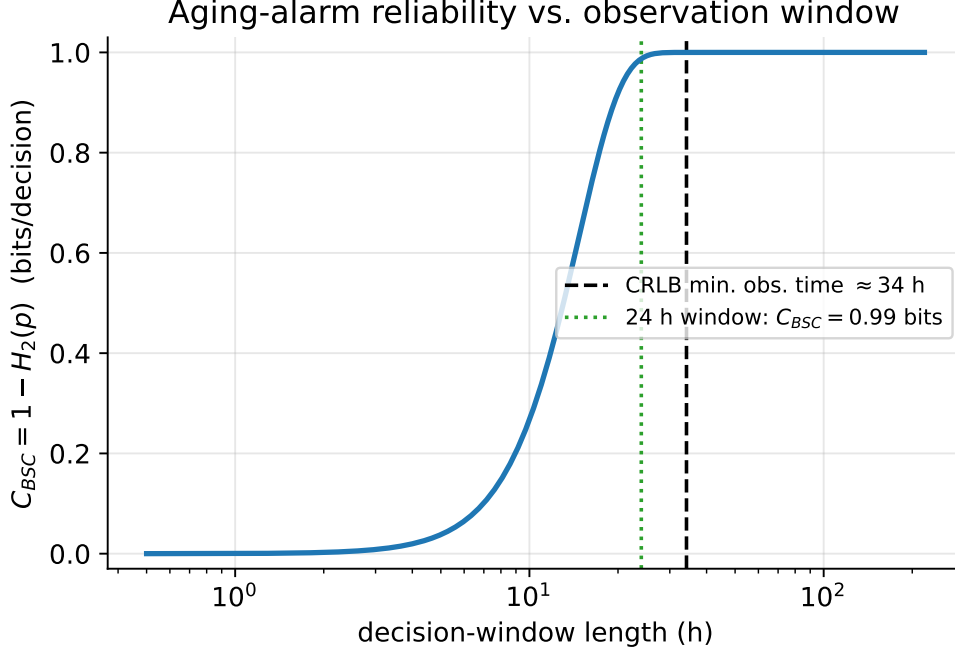


Figure 4: Aging-alarm channel capacity $C_{BSC} = 1 - H_2(p)$ as a function of the decision-window length. Reliability climbs from ≈ 0 (per-hour, buried in noise) to ≈ 1 bit, with the transition coinciding with the Cramér–Rao minimum observation time (≈ 34 h, dashed)—the three figures of merit are mutually consistent.

correction without using the measured T, V coupling. A second component aligns with temperature ($|\text{corr}(\cdot, T)| = 0.688$). Both recovered sources are strongly non-Gaussian (excess kurtosis $+5.82$ and $+29.91$), confirming the premise that non-Gaussianity is the feature distinguishing the physical sources—and consistent with the heavy-tailed excursions the KL analysis (section 6.3) exposed.

The separation is *partial*, and is reported as such: the residual pairwise mutual information between recovered components reaches 0.571 bits (ideal independence is 0), because the slow, non-stationary aging drift and the heavy-tailed noise only approximately satisfy ICA’s i.i.d./stationarity assumptions. ICA is therefore corroborative here—an independent, assumption-light route to the same aging component—rather than a clean unmixing. That a method using none of the T, V coupling still recovers 83% of the regression aging signal is itself the result.

7 Discussion

7.1 What information theory adds over R^2

A correct, transferable noise metric: mutual information is valid where the physics is non-linear and is expressed in bits. On the (PID) aging device the measured dependence exceeds the linear- R^2 equivalent by 26 %, showing that R^2 systematically understates the true environment–sensor coupling. This *implies* that the prior R^2 -based comparison of thermal regimes understates their difference; recomputing both benches head-to-head in bits would require a matched run on the same device, which is outside this single-regime study (section 7, threats to validity).

7.2 A principled definition of burn-in quality

Reframing burn-in quality as the minimization of $I(S; T, V)$ gives the qualitative claim “PID is a cleaner validation environment” a rigorous unit and an optimization target: every bit the

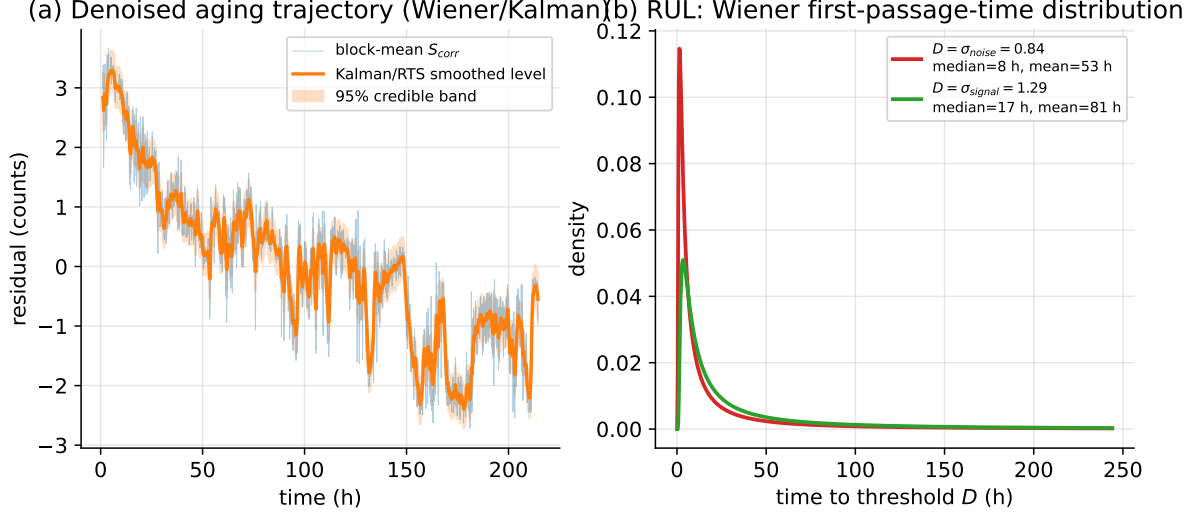


Figure 5: (a) Block-mean S_{corr} (light) and the Kalman/RTS-smoothed aging trajectory $\hat{X}(t)$ with a 95% credible band. (b) The Inverse-Gaussian RUL distribution (eq. (14)) at two Tier-1-anchored thresholds: one noise-floor increment ($D = \sigma_{noise}$) and one already-observed signal-scale excursion ($D = \sigma_{signal}$).

environment writes into the slack reading is a bit of aging information lost. We state this as a principle rather than demonstrate it within one device; the closest empirical support here is the distributional gap between the two benches (section 6.3), measured on a separate test device.

7.3 A description-length view of the linear-vs-non-linear choice

The MDL criterion of section 3.6 that selects the linear PVT correction (section 6.2) is not an ad-hoc penalty on extra parameters: it is a computable stand-in for the Kolmogorov-complexity intuition that the best model is the *shortest description* that still accounts for the data (eq. (9)). Framed this way, the result is not merely “the non-linear correction is not worth it here”—it is that at this single, narrow operating point (125.1 °C, 1.400 V) the data do not yet contain enough curvature information to justify paying for the Arrhenius/super-linear model’s extra parameters, even though the underlying physics is genuinely non-linear (section 2.5). This is consistent with, not contradictory to, the non-linearity result of section 6.1: MI detects non-linear *dependence* that R^2 misses, while MDL asks whether a specific parametric non-linear *form* pays for itself in *this* narrow-range dataset. A wider-range or multi-setpoint campaign, spanning enough of the Arrhenius/BTI curvature to be compressible by the non-linear model, is the natural condition under which MDL would flip its verdict.

7.4 A resolution limit for SLM sensors

The capacity and BSC results put concrete numbers on what the sensor can do: 1.8 distinguishable aging states over the campaign, a daily (24 h) aging alarm of $C_{BSC} = 0.99$ bits, and a 3σ detection floor of 1.2 m-cnt/h reached after ~ 34 h. These tell an SLM designer how finely the sensor resolves aging and how reliably it can raise an alarm under realistic noise—quantities directly consumed by adaptive voltage scaling and RUL, and which the prior residual- σ metric left implicit. Their mutual consistency (the BSC per-window reliability crosses over near the CRLB minimum observation time) is evidence the channel model is coherent rather than three disconnected heuristics.

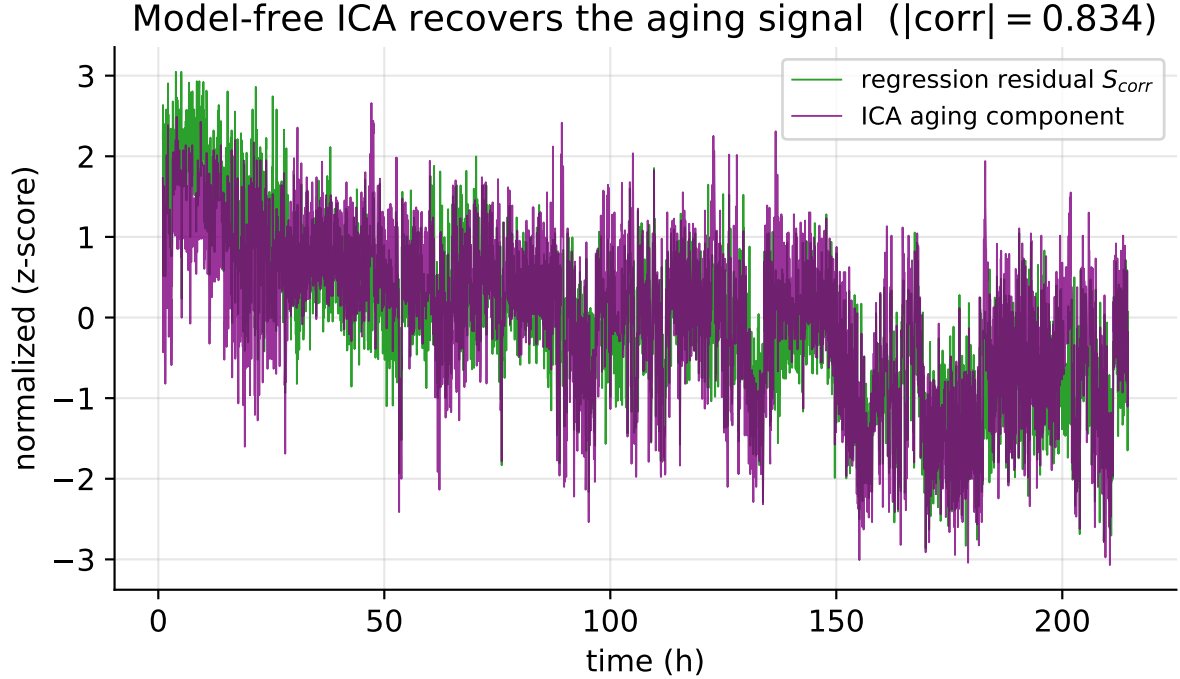


Figure 6: The ICA-recovered aging component (purple) overlaid on the regression-based residual S_{corr} (green), both z -scored. Model-free ICA, given only (S, T, V) , tracks the regression aging signal at $|\text{corr}| = 0.834$.

7.5 Sequential estimation and RUL as a distribution

The Wiener/Kalman treatment of section 6.6 is deliberately not a replacement for the Cramér–Rao/Bayesian bound of section 6.5, but a cross-check from an independent estimation route (a dynamic, filtered model rather than a single whole-series regression) that happens to agree on the sign and order of magnitude of the aging rate. Its added value is structural, not numerical: it turns a single point-rate-with-error-bar into (i) a continuously updated, denoised trajectory estimate the operator could in principle consume online, and (ii) a full RUL *distribution* rather than a bound, exposing the right-skew that a Gaussian error bar hides. That the rate-noise MLE and the MDL criterion of section 6.2 *independently* both prefer the simplest available model (constant drift; linear correction) at this single operating point is itself a finding: two different estimation principles, applied to two different questions, agree that this campaign does not yet contain the curvature needed to justify a more flexible model — exactly the kind of cross-validation this project has used throughout (KSG vs. binned MI, frequentist vs. Bayesian RUL, ICA vs. regression correction).

7.6 Threats to validity

The aging-device analysis is a single device and single thermal regime, which limits permanent-aging separability. The bench comparison of section 6.3 is on a *different* test device, so its KL/JS divergence speaks to bench fidelity, not to the aging device’s behaviour, and the two are never mixed. MI estimates on autocorrelated data are bias-prone despite the large raw sample count, mitigated (not eliminated) by n_{eff} , the surrogate null floor of section 6.1 and cross-estimator agreement; the capacity expression is an effective-resolution heuristic; the ICA separation is corroborative, not a clean unmixing. The Wiener/Kalman model of section 6.6 assumes constant block-mean observation noise and a genuinely diffusive (Wiener) latent process; the RUL distribution it produces is reported at self-referential, already-computed thresholds

precisely because no independently calibrated failure threshold exists for this sensor, so it is a *methodology* demonstration, not an operational lifetime prediction, in the same spirit as the Cramér–Rao bound of section 6.5.

8 Conclusion and Future Work

We placed an information-theoretic measurement layer on an existing accelerated-aging burn-in platform, re-expressing environmental contamination, PVT correction and aging recoverability as entropy, mutual information and channel capacity. On a 214.6 h PID campaign we established the empirical anchors ($H(S) = 3.43$ bits; linear $R^2 = 0.738$) and a methodology that replaces the model-dependent R^2 with mutual information, defines burn-in quality as the minimization of $I(S; T, V)$, converts measured noise into resolvable aging states and an alarm-channel capacity, and bounds RUL detectability honestly via Cramér–Rao and Bayesian estimation. A Wiener-process/Kalman-filter extension of the same estimation toolbox further turns this single-point RUL bound into a continuously updated, denoised aging trajectory and a closed-form Inverse-Gaussian RUL *distribution* (section 6.6), cross-validated against the Cramér–Rao/Bayesian estimate and against the MDL model-selection result of section 6.2.

Future work. Acquire a matched-length bang-bang campaign to complete the cross-regime KL and capacity comparison on fresh data; extend to multiple devices to separate a permanent-aging trajectory from device-to-device variation; deploy the windowed pipeline online as a continuous information-feature generator feeding the SLM controller, with the Kalman-smoothed trajectory of section 6.6 as its live aging estimate; calibrate an actual failure threshold D (e.g. from a timing-margin specification) so eq. (14) yields an operational RUL distribution rather than the self-referential thresholds used here; and pursue the ICA/blind-separation route with additional derived channels to strengthen model-free aging recovery.

Reproducibility

Every scalar in this paper is generated by the analysis pipeline into `generated/values.tex` and every figure/table into `figures/` and `tables/`; re-running the analysis on a new log regenerates the document without manual transcription.

Acknowledgments

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